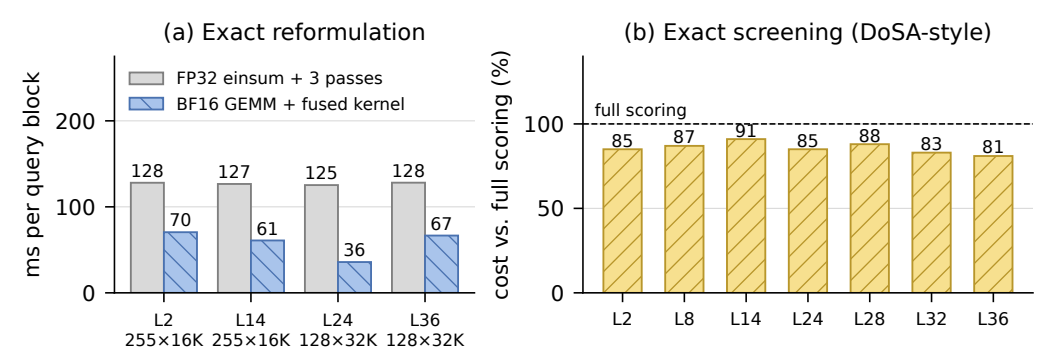
(a) Exact reformulation
FP32 einsum + 3 passes
BF16 GEMM + fused kernel
ms per query block
0
100
200
128
70
127
61
125
36
128
67
L2
255×16K
L14
255×16K
L24
128×32K
L36
128×32K
(b) Exact screening (DoSA-style)
cost vs. full scoring (%)
full scoring
0
50
100
85
87
91
85
88
83
81
L2
L8
L14
L24
L28
L32
L36